

SAI AKSHAYA N

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OBJECTIVE

Electronics and Communication Engineering candidate with a strong interest in core electronics, embedded systems, and IoT devices, seeking hands-on exposure in device installation, testing, servicing, and field-oriented engineering roles.

EDUCATION

B.E. in Electronics and Communication Engineering

2022 – Present

Atria Institute of Technology, Bangalore

CGPA: 8.01

Relevant Coursework: Digital Electronics, Microcontrollers, Embedded Systems, VLSI Basics and IoT

PROJECTS

AI-Powered Smart Dispenser With Computer Vision Based Volume Estimation

- Designed an AI-IoT-enabled dispenser that uses computer vision to detect container volume and dispense water precisely.
- Integrated Raspberry Pi and sensors for automated volume estimation and flow control, reducing water consumption by 12%.

Amphibious Screw-Propelled Helix Robot

- Developed an ESP32-S3 based mobile robot capable of land–water navigation achieving stable traversal across three different terrains.
- Programmed embedded control for sensor data acquisition, motor driving, increasing multi-terrain motion stability by 15%.

INTERNSHIP EXPERIENCE

Robotics & System Intelligence Intern

Feb 2026 – Present

ERA Foundation, India

- Contributed to development of embedded, IoT, and vision-based systems using Raspberry Pi, ESP8266, Python, and OpenCV
- Implemented sensor interfacing and GPIO-based control for motors, relays, cutting response latency by 25% in real-time monitoring applications.
- Developed HSV-based vision algorithms for line detection and color recognition achieving ~90% accuracy
- Collaborated with team members to design, integrate, and test prototypes for automation systems

TECHNICAL SKILLS

Programming Languages: C/C++, Python, MATLAB

Tools & Platforms: Cadence, Arduino, Keil μ Vision 4

Communication Protocols: SPI, UART, I2C

Version Control: Git, GitHub

CERTIFICATIONS

Analog Circuits – NPTEL

Linux Operating Systems – Udemy

Virtual Internship – Embedded Systems, Microchip Technology

Certificate of Presentation – ICSIE 2025: Presented “*Computer Vision Based Volume Estimation*”, Prince Dr. K. Vasudevan College of Engineering & Technology (Oct 2025)